

Milestone 3 — Security, Testing, Audit Trail, and Deployment

Goal

Enhance the airline reservation system by implementing application security, automated testing, audit trail logging, and deployment. By the end of this milestone, the application should be secure, tested, and accessible through the CCS Cloud Server (CCSCloud).

Learning Objectives

Students should demonstrate the ability to:

- Implement session-based authentication.
 - Secure application pages using middleware.
 - Apply Role-Based Access Control (RBAC).
 - Write unit tests using Jest.
 - Record user activities using an audit trail.
 - Deploy a Node.js web application to CCSCloud.
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1. Session Management

Implement session-based authentication using Express sessions.

Requirements

- Users must log in before accessing protected pages.
- Store authenticated user information in a server-side session.
- Destroy the session upon logout.
- Redirect unauthenticated users to the login page when accessing protected pages.

Protected Pages

The following pages must only be accessible to authenticated users:

- Profile
- Flight Booking

- My Reservations
 - Flight Management
 - Reservation Management
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2. Role-Based Access Control (RBAC)

The system must support two user roles:

- Administrator
- Passenger

Access to pages and system features must be controlled according to the user's assigned role.

Passenger Permissions

Passengers may:

- View and update their profile
- Search available flights
- Book flights
- View their own reservations
- Cancel their own reservations

Passengers must not be able to:

- Access administrator pages
- Manage flights
- View reservations belonging to other users

Administrator Permissions

Administrators may:

- Manage flights
- View all reservations
- Manage user accounts
- View audit trail logs

Administrators are not required to perform passenger booking transactions.

Middleware

Implement middleware to:

- Verify that the user is authenticated.
 - Verify that the user has the required role before accessing protected routes.
 - Redirect unauthorized users to an appropriate page or display an access denied message.
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3. Unit Testing

Use **Jest** to create automated unit tests.

Required Test Cases

User Authentication

- Successful registration
- Successful login
- Failed login

Flight Management

- Create Flight
- Update Flight
- Delete Flight

Reservation Management

- Create Reservation
- Cancel Reservation

Business Rule Validation

Test at least one invalid scenario, such as:

- Booking a flight with no available seats
- Selecting an occupied seat

Include both successful and unsuccessful test cases where applicable.

4. Audit Trail Logging

Implement an audit trail to record important user activities.

Activities to Record

- User Registration
- User Login
- User Logout
- Flight Creation
- Flight Update
- Flight Deletion
- Reservation Creation
- Reservation Cancellation

Audit Log Information

Each audit log entry should include:

- Date and Time
- Username
- User Role
- Activity Performed

Audit trail records should be stored in a MongoDB collection.

Audit Log Viewer

Administrators should be able to view the audit trail through an audit log page.

5. Deployment to CCSCloud

Deploy the completed application to the **CCS Cloud Server (CCSCloud)**.

Deployment Requirements

Students must:

- Deploy the Node.js application to CCSCloud.
- Configure the application to connect to the production MongoDB database.

- Ensure all pages and features are functional after deployment.
- Verify that session management works correctly in the deployed environment.
- Verify that RBAC restrictions remain enforced after deployment.
- Verify that audit trail logging continues to record user activities after deployment.

Deployment Verification

The deployed application should allow the instructor to:

- Access the application through the deployment URL.
- Log in using administrator and passenger accounts.
- Access protected pages after authentication.
- Verify administrator and passenger permissions.
- Manage flights.
- Create and cancel reservations.
- View audit trail logs as an administrator.

Students should thoroughly test the deployed application before submission.

Deliverables

Security

- Session-based authentication
- Protected routes using middleware
- Role-Based Access Control (RBAC)

Testing

- Jest test files
- Test execution results

Audit Trail

- Audit log collection in MongoDB
- Audit log viewer for administrators

Deployment

- Successfully deployed application on CCSCloud
- Deployment URL

- Administrator test account credentials
- Passenger test account credentials

Documentation

Submit the following documentation:

- RBAC Permission Matrix
- List of Protected Routes
- Unit Test Summary
- Audit Trail Sample Output
- Deployment URL

Milestone 3 Deliverables Summary

By the end of Milestone 3, the completed application should:

1. Require users to authenticate before accessing protected pages.
2. Restrict access to system features using Role-Based Access Control (RBAC).
3. Protect application routes using authentication and authorization middleware.
4. Include automated unit tests for major application features using Jest.
5. Record important user activities through an audit trail.
6. Allow administrators to view audit logs.
7. Be successfully deployed and fully functional on CCSCloud.